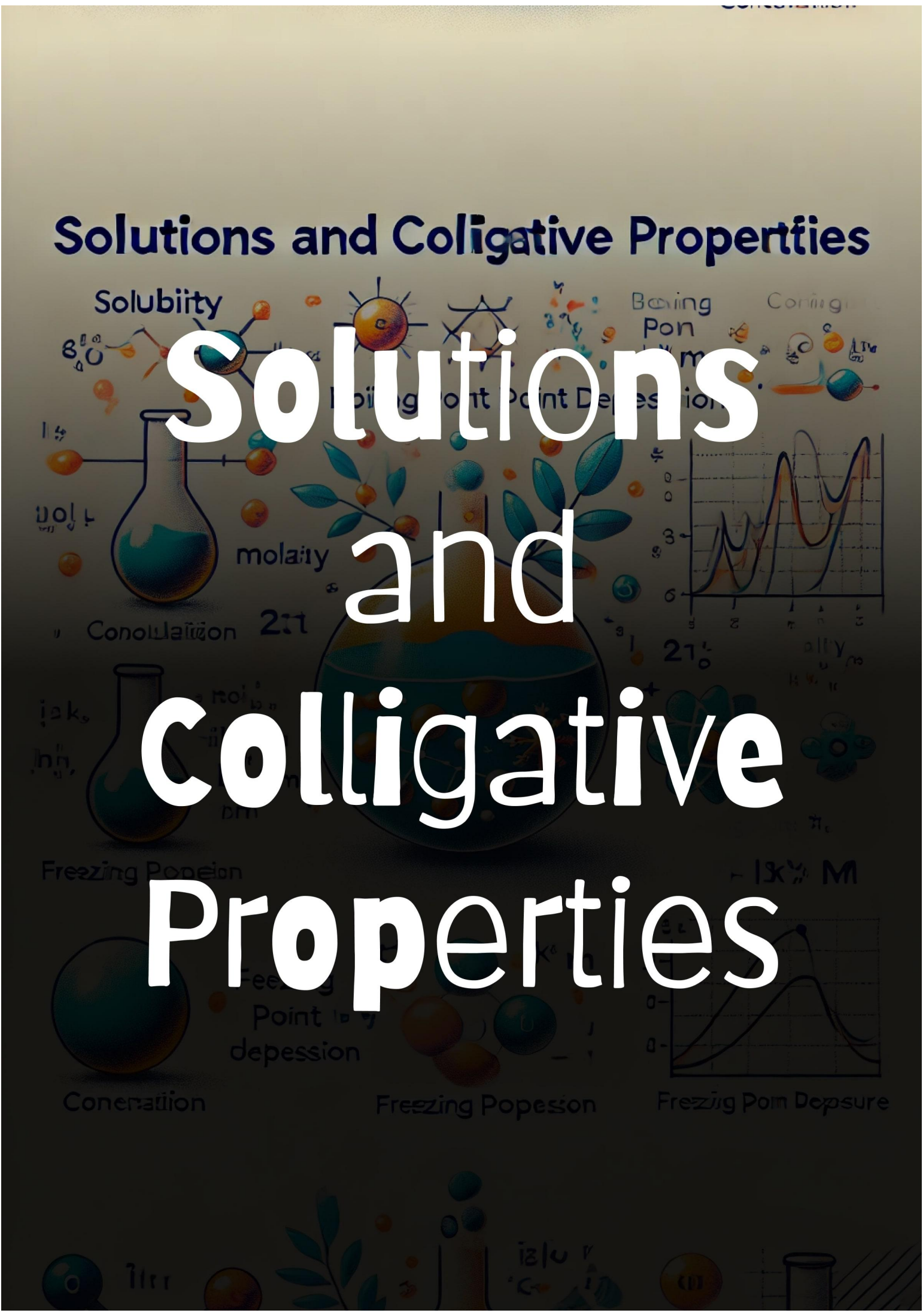


Solutions and Colligative Properties

SOLUTION AND COLLIGATIVE PROPERTIES

Solutions have two types of properties :-

a. **Constitutional Properties** – These depend on the nature of solute.

b. **Colligate Properties** – These depend on the number of solute particles.

These are 4 types of colligate properties.

1. Dramatic pressure
2. RLVP $\left(\frac{P^\circ - P_\Delta}{P^s}\right)$
3. Elevation in BP
4. Depression in freezing point (ΔT_F)

VANT HALF FACTER

(i) It is observed colligative property Theoretical.

a. "i" for dissociation.

$An \rightleftharpoons nA$

$$i = \frac{\alpha(1-\alpha) + n\alpha\alpha}{\alpha}$$

$t = a$

$t = ta(-\alpha+1) \quad na\alpha$

$i = 1 + (n-1)\alpha, \quad i > 1$

b. "i" for association

$nA \rightleftharpoons An$

$$i = \frac{a\beta\left(-\frac{1}{n}+1\right) + a}{a}$$

$t = 0 \quad a$

$t = t \quad a(1-\beta) \quad \frac{a\beta}{n}$

$i = 1 + \beta\left(\frac{1}{n} - 1\right), \quad i < 1$

$n \geq 2$

If solute will neither associate nor dissociate, then $(i = 1)$

eg:- sucrose, urea.

OSMOSIS

When two liquids having different conc^n are seperated by semi-permeable membrane, then there will be spontaneous movement of particles from low conc^n to high conc^n. This process is called "osmosis."

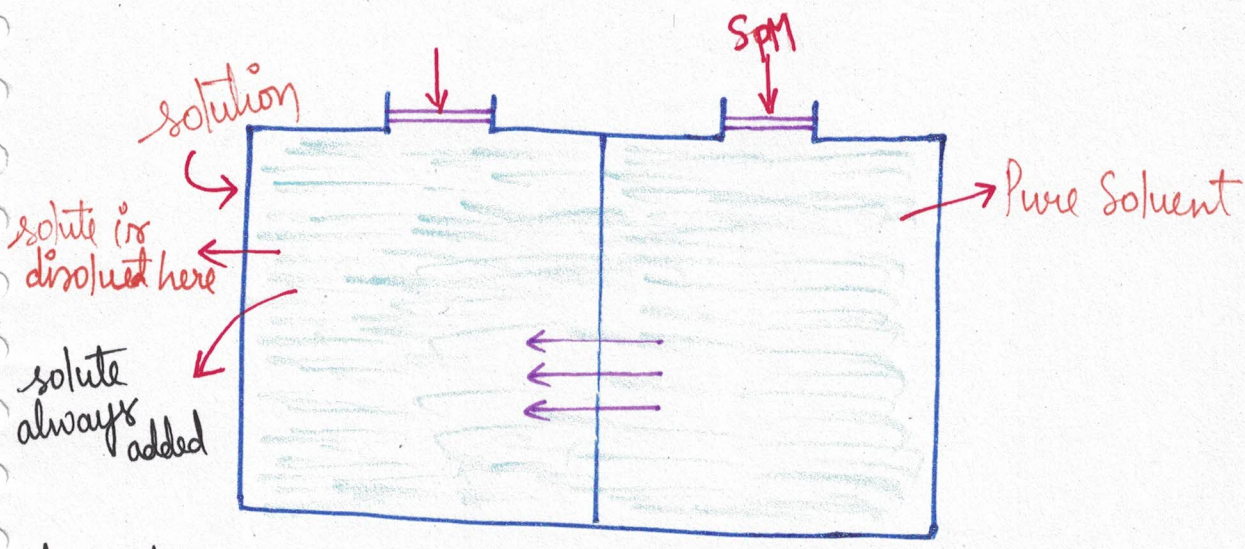
The pressure applied that can stop osmosis is called osmotic pressure.

If osmotic pressure is "π", then it is proportional to the conc^{n} & temperature.

$\pi = iCRT$

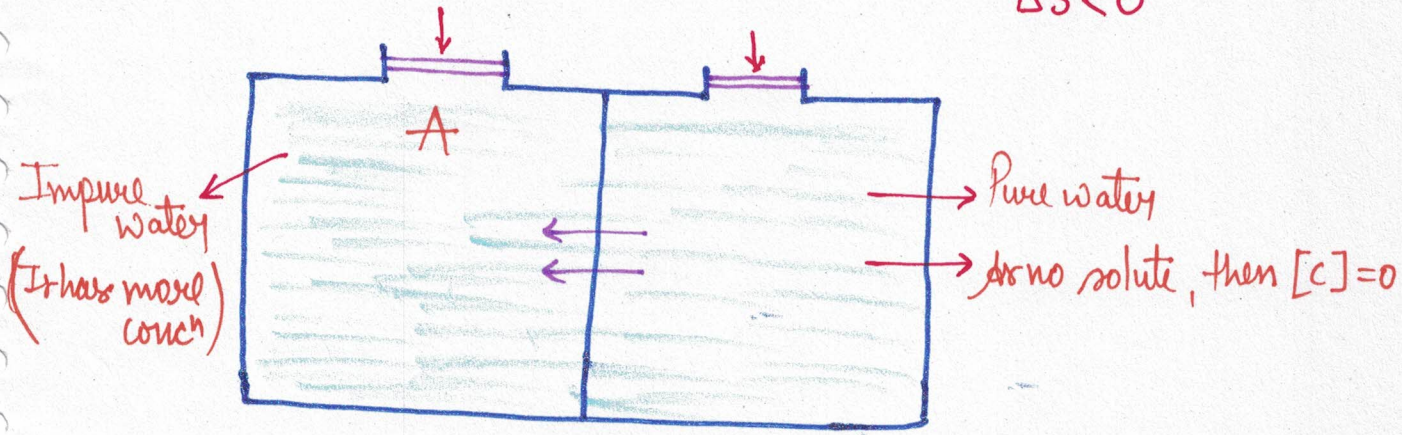
$\pi \propto C$ $\quad\quad$ $\pi = SCT$

$\pi \propto T$ $\quad\quad$ $S = 0.0821$ lt atm/mole K



Some permeable membrance allows only solvent particle.

$\Delta S < 0$



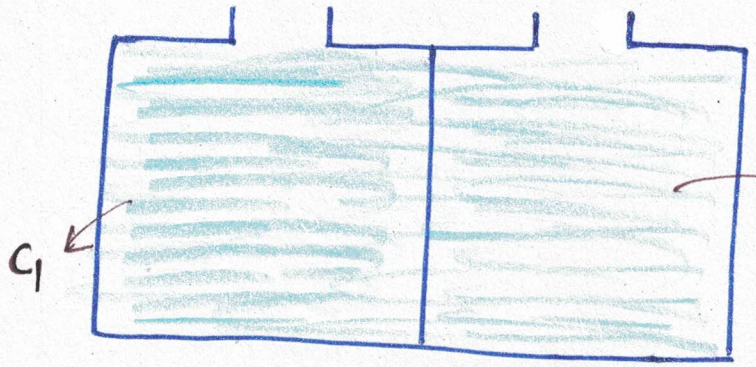
- Thus the Natural process says pure water moves towards impure water. Thus, pressure has to be applied on container A.

- Natural SPM = Animal membrance
  Artificial SPM = Silicates of Fe, Co, Ni

REVERSE OSMOSIS

- When we'll apply pressure more than osmotic pressure, then

Osmosis will be reversed and the process is called reverse osmosis (Rs).



(If solute is added in both)

$(\pi)_{Net} = \pi_1 - \pi_2$

$i C_1 RT = i C_2 RT$

If $\pi_1 = \pi_2 \rightarrow$ Isotonic solution

If $\pi_1 > \pi_2 \rightarrow$ solution 1 is hypertonic with soln 2.

? What will be "π" of 0.1M urea solution at 27°C

(i) $\pi = (1)\left(\frac{1}{10}\right)(0.08)(300) = 2.4$ atm $= (\int gh)$

As temp = K thus $P_1V_1 = P_2V_2$

(ii) "π" of 0.1M NaCl soln at 27°C (aq.)

$\pi = (2)\left(\frac{1}{10}\right)\left(\frac{1}{12}\right)(300) = 4.8$ atm

(iii) "π" of 0.1M $K_4[Fe(CN)_6]$ $\alpha = 50\%$, $t = 27°C$

$$\underset{t=t}{K_4[Fe(N)_6]} \longrightarrow \underset{4\alpha}{4K^+} + \underset{\alpha}{[Fe(CN)_6]^{-4}}$$

$\underset{1-\alpha}{}$

$n_T = 1 + 4\alpha = 3$ $\quad \pi = (3)(0.1)\left(\frac{1}{12}\right)(300) = 7.2$ atm

(iv) "π" of soln containing 300ml 0.2M urea + 200ml 0.4M NaCl + 500ml 0.1M K_2SO_4 at 27°C

$n_1 = 300 \times 0.2 = 60$ millimoles

$n_2 = 200 \times 0.4 \times 2 = 160$ millimoles

$n_3 = 500 \times 0.1 \times 3 = 150$ millimoles

$[C] = \frac{60 + 160 + 150}{1000} = \left(\frac{37}{100}\right)$

$(\pi)_{soln} = \left(\frac{37}{100}\right)\left(\frac{1}{12}\right)_4(300) = \frac{37}{4}$ atm.

Colligative Property $\propto \frac{1}{(\text{Mass})_{\text{solute}}}$

VAPOUR PRESSURE

It is pressure executed by vapours on its Liquid when liquid is in eqm with vapours.

$VP \propto T \longrightarrow$ only

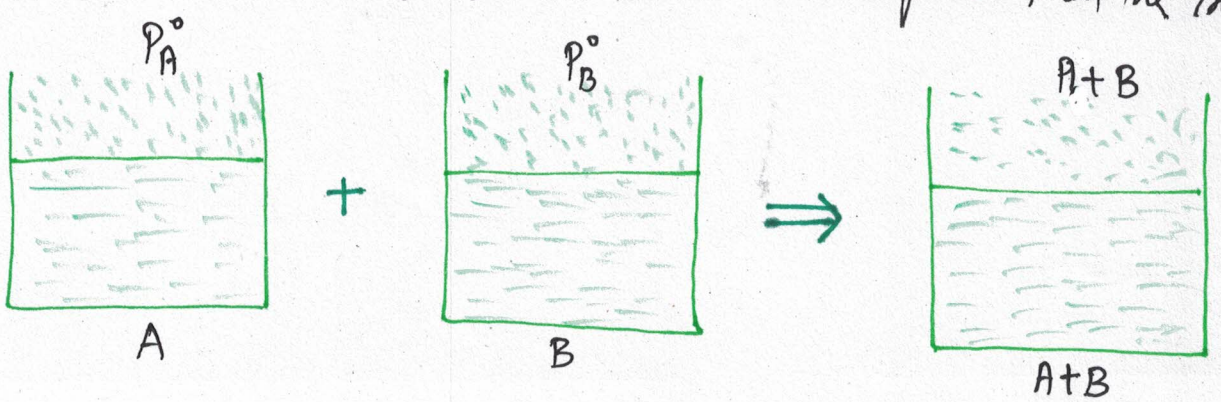
$$\log\left(\frac{P_2}{P_1}\right) = \frac{\Delta H}{(2.303)R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

If attraction force between molecules are more, then VP is Low.

VAPOUR PRESSURE OF SOLUTION

RAOULT'S LAW

V.P of any liquid will be proportional to its mole fraction in the solution.



P_A° = VP of pure A; P_B° = V.P of pure B

$P_A \propto x_B \longrightarrow P_A = K x_A$, If $x_A = 1 \longrightarrow P_A = P_B^\circ$

$P_B \propto x_B \longrightarrow P_A = x_A P_A^\circ$, $P_B = x_B P_B^\circ$

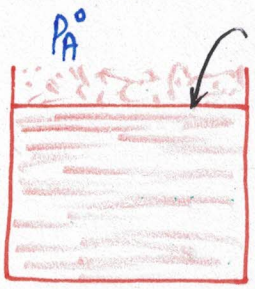
$$P_{Total} = P_A + P_B$$

$$P_{solution} = x_A P_A^\circ + x_B P_B^\circ$$

RELATIVE LOWERING IN VAPOUR PRESSURE (RLVP)

When a non volatile solute is added to a volatile solvent, then Vapour Pressure will decrease. Always will multiply by

at the answer



$P_B^\circ = 0$

$P_S = x_A P_A^\circ + x_B P_B^\circ$

when B is solute, $P_B^\circ = 0$

$P_S = x_A P_A^\circ$

$P_S = (1 - x_B)(P_A^\circ) \Rightarrow$ $\boxed{i x_B = \frac{P_A^\circ - P_S}{P_A^\circ}}$

$$\frac{P^\circ - P_S}{P^\circ} = \frac{i\, n_{solute}}{n_{solute} + n_{solvent}}$$

Also, $$-1 + \frac{P^\circ}{P^\circ - P_S} = \frac{n_{solute} + n_{solvent} + (-1)}{n_{solute}}$$

$$\frac{P^\circ - P_s}{P_s} = \frac{(i)\, n_{solute}}{n_{solvent}}$$

P_s = solution

P° = Pure A

When the solution is very dilute $\Rightarrow n_{solvent} > n_{solute}$

$$\frac{P^\circ - P_s}{P^\circ} = i\,\frac{n_{solute}}{n_{solvent}}$$

$P^\circ - P_s$ = Lowering in V.P., $\frac{P^\circ - P_s}{P^\circ}$ = RLVP

Sup. – VP of ether at 20°C = 442 mm. When 7.2 g of a solute is dissolved in 60 g ether, the VP is lowered by 32 units. Calculate molecular weight of solute if m_{ether} = 74 g

Ans. – $(VP)_{ether} = 442$ mm, $T = 293$ K

$(P^\circ)_A = 442$ mm, $n_{solute} = \frac{7.2}{M}$, $n_{solvent} = \frac{\cancel{60}\,30}{\cancel{74}\,37}$

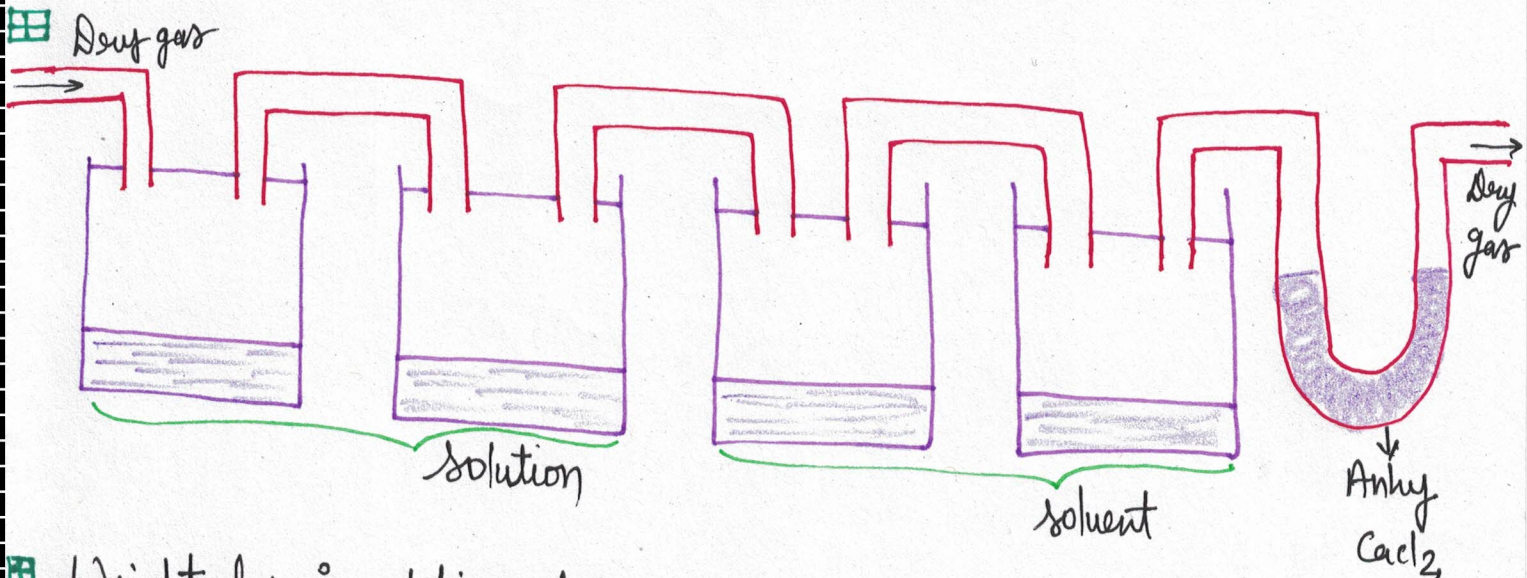
$(P)_\Delta = 442 - 32 = 410$ mm

$$\therefore \frac{442-410}{410} = \frac{\left(\frac{72}{10M}\right)}{\left(\frac{30}{37}\right)} = \frac{72\times 37}{300M} \Rightarrow M = \frac{37\times 37\times 41}{8\times 60}\ g.$$

$$\frac{\cancel{32}^{\,8}}{\cancel{410}\ \cancel{205}\ 41} = \frac{\cancel{72}^{\,37}\times 37}{\cancel{300}\,M\ 60}$$

$M = 113.77$ g Ans.

Experimental Method to determine RLVP (Ostwald Walker Method)



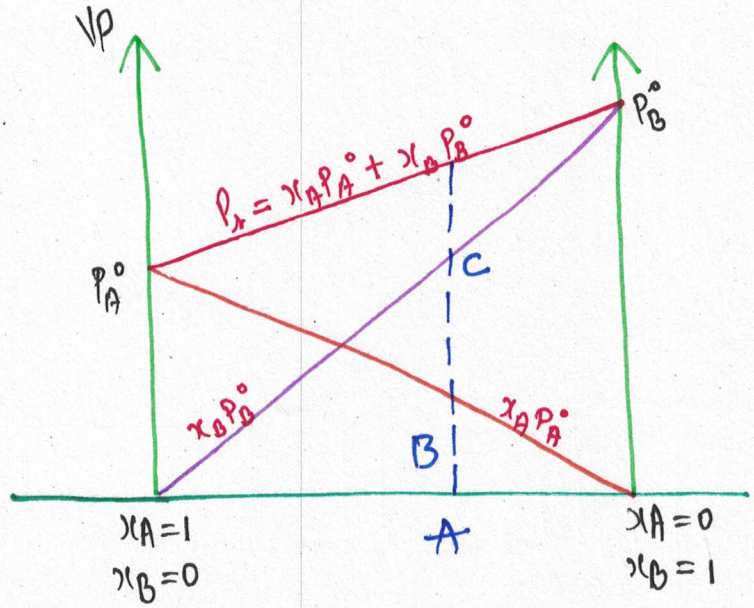
- Weight loss in solution container $\propto P_s$
  Weight loss in solvent container $\propto P^\circ - P_s$
  Weight gain in $CaCl_2$ container $\propto 1^\circ - P_t + P_s$
- If the solvent container would have been earlier $\propto P_0$
  Weight loss in solvent container $\propto P^\circ$
  Weight loss in solution container $\propto P^\circ + P_s$
  Weight gain in $CaCl_2$ container $\propto P^\circ$

Ideal Solution

The solution in which $\Delta H_{mixing} = 0$ $\Delta V_{mixing} = 0$. They follow Raoult's Law.

If the forces A-A, B-B, A-B are same

$$P_s = x_A P_A^\circ + x_B P_B^\circ$$



AB + AC = ?

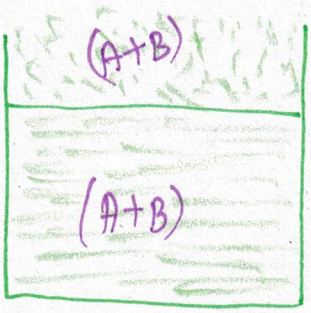
↓ ↳

$x_A P_A^\circ$ $x_B P_B^\circ$

AB + AC = AD

VP will be between P_A° and P_B°.

More Volatile component is always costlier.



$$P_A = y_A P_s \Rightarrow y_A = \frac{x_A P_A^\circ}{x_A P_A^\circ + x_B P_B^\circ}$$

$$P_B = y_B P_s \Rightarrow y_B = \frac{x_B P_B^\circ}{x_A P_A^\circ + x_B P_B^\circ}$$

$y_A \Rightarrow$ Vapour composition of A

$y_B \Rightarrow$ Vapour composition of B

Q. V.P of pure "A" = 100 mm of Hg | Calculate :-

V.P of pure "B" = 150 mm of Hg | Calculate :-

Q. $(V.P)_{Total}$ & Vapour composition if an liquid form 3 moles of A and 2 mole of B present.

Ans :- @ $x_A = \frac{3}{5}$, $x_B = \frac{2}{5}$ $\quad P_s = \left(\frac{1}{5}\right)[3 \times 100 + 2 \times 150]$

$P_s = 120$ mm.

$y_A = \frac{3/5 \times 100}{120} = 0.5$; $y_B = 0.5$

(b) If condensed, they find new mole fraction and the new $(V.P)_{Total}$

Ans. $(x_A = 1/2 , x_B = 1/2) \longleftarrow (x_A)_i^\circ = 3/5 , (x_B)_i^\circ = \frac{2}{5}$

Thus, B gets enriched, it had more V.P. Thus, doing many times, we can get pure (B) → Fractional Distillation.

$$(VP)_s = 1/2(100) + 1/2(150) = 125$$

(c) Now, find new vap. composn of A and B.

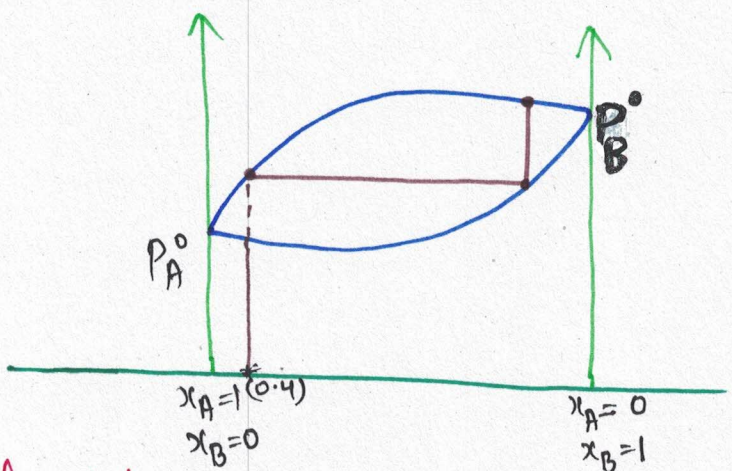
Ans.

$$(Y_A)_{new} = \frac{1/2(100)}{125} = 0.4$$

$$(Y_B)_{new} = \frac{75}{125} = 0.6$$

Thus, similarly we can separate it.

The last question shows the process of distillation.



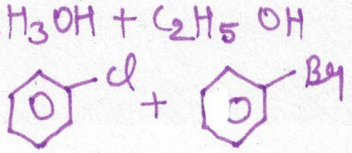
e.g of Ideal solution

a. Benzene Toulene

b. Methyl bromide + ethyle iodide

c. Butyl chloride + Buthyl bromide

d. $CH_3OH + C_2H_5OH$

e. 

NON IDEAL SOLUTION

Those solution which don't follow Raoult's Law. It is of two kinds :-

(i) Positive deviation

$P_A > x_A P_A^\circ$, $P_B > x_B P_B^\circ$

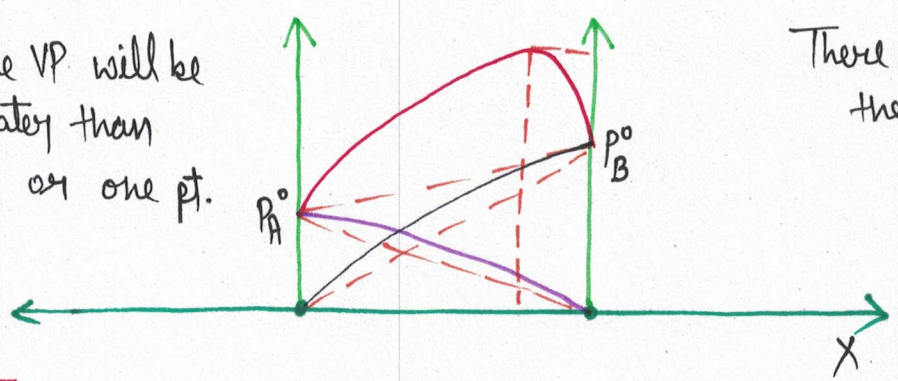
$P_s > P_A^\circ x_A + x_B P_B^\circ$

$\Delta H_{mix} > 0$, $\Delta V_{mixing} > 0$.

Thus the mixture will be colder ($\Delta H_{mix} > 0$).

A-B < (A-A and B-B)

The VP will be greater than P_B° at one pt.



There will be one point where the VP will be $> P_B^\circ$

e.g.:-

a. Acetone + ethanol

b. Acetone + CS_2

c. Acetone + Benzene

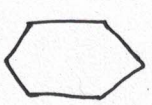
d. CCl_4 + Toulene

e. CCl_4 + chloroform

f. $CCl_4 + CH_3OH$

g. $H_2O + C_2H_5OH$

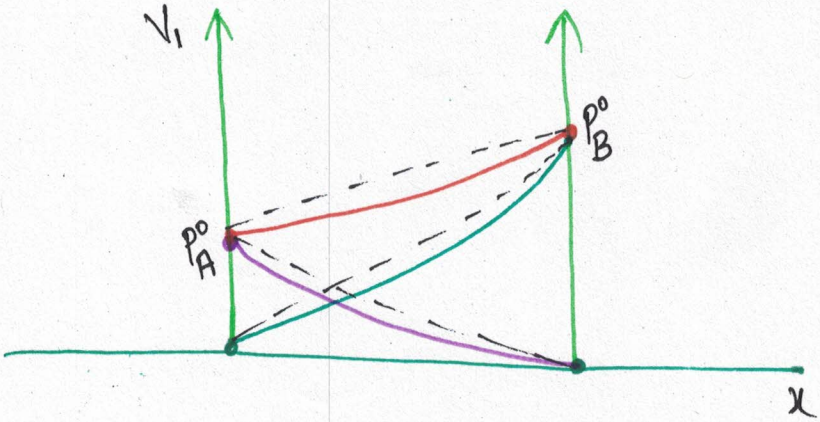
h. $H_2O + CH_3OH$

i.  $+ C_2H_5OH$

Negative deviation

$P_A < x_A P_A^\circ$, $P_B < x_B P_B^\circ$, $P_s < x_A P_A^\circ + x_B P_B^\circ$

$\Delta H_{mix} < 0$, $\Delta V_{mix} < 0$, (A--B) > (A--A, B--B)

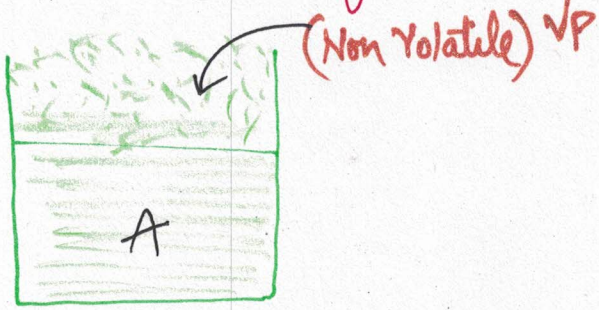


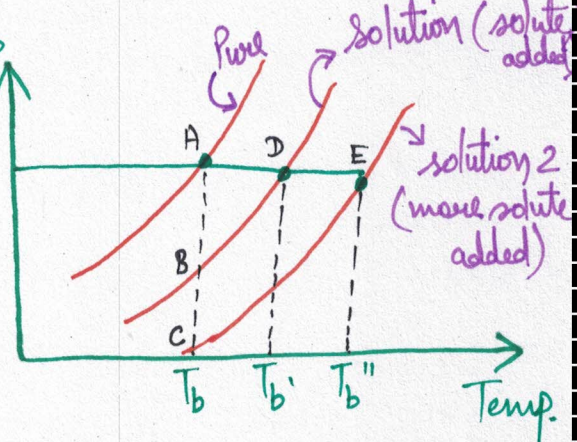
eg:-

a. Acetone + aniline
b. Acetic acid + Pyridine
c. Acetic acid + CH_3OH
d. $CHCl_3$ + Et - O - Et
e. CH_3Cl_3 + Benzene
f. H_2O + HCl
g. HNO_3 + H_2O

Elevation in Boiling Point

It is the temperature at which the V.P of any liquid is equal to the external atmospheric pressure. If atmospheric pressure is 1 atm, then it is called "normal boiling point."





On adding solute, the VP of the solution ↓

$\Delta P \propto \Delta T_b$ $\Delta p \propto$ molality

$\Delta T_b = i K_b \text{ Molality}$

units of K_b (Molal elevation constant)
$= K \text{ kg/mol}$.

$K_b \rightarrow$ ebullioscopic constant

$l_v \rightarrow$ latent heat of vapour

Experimentally; $K_b = \frac{RT_b^2}{1000 l_v}$

$R = 8.31$ J/mole K or 2 Cal/mole K

$T_b =$ Normal boiling point of solvent

For Water - $K_b = \frac{(2)(373)^2}{(1000)(540)} = \frac{373 \times 373 \times 2}{54 \times 104} = 0.515$

$\left(K_b\right)_{water} = 0.52 \text{ K kg/mol}$

Q. A Solution of 1g phenol in 50 ml of diethyl ether boiled at a temp. elevated by 0.632°C. The Molal elevation constant for diethyl ether is 2.12 K mol 1 kg. Density is 0.714 g/ml. Calculate molecular formula of phenol.

$\Delta T_b = 1 K_b \quad \text{Molality} = (0.632) = (2.12)M$

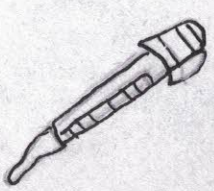
Mass/phenol $\frac{(0.632)}{(2.12)} \times \frac{(50)(0.714)}{(1000)} =$ ✓

Mass of Solvent $= (50)(0.714)$

Q. A solution of 1.21g of a solute ($M = 152$) in 26.7g acetone bais at 329.8K. Calculate (l_v) per mole of acetone. Pure acetone boils at 329.3K.

A. $\Delta T_b = 0.5K \quad K_b = \frac{(2)(329.8)^2}{(1000)(l_v)}$

$(0.5) = \frac{(2)(329.8)^2}{(1000)(l_v)} \cdot \frac{(1.21)}{(152)} \frac{(1000)}{(27.7)}$



Here, "l_V" we get in (cal/gram). Then, get "l_m" in (cal/mole)

Depression in Freezing Point

$l_f = 80$ cal/gm

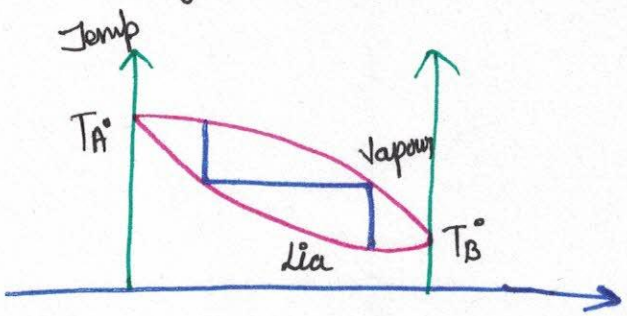
K_f = molal depression constant or cryoscopic constant.

$$K_f = \frac{R(T_f)^2}{1000 l_f}, \quad K_f = 1.86 \text{ k kg/mol}$$

$$\Delta T_f = i k_f \text{ molality}$$

DISTILATION

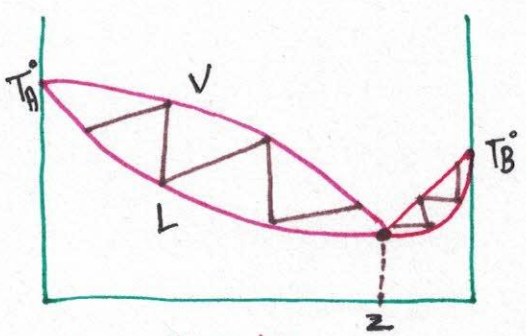
It is a process to separate two or more than two liquids having different Boiling points.



$VP \propto \frac{1}{BP}$

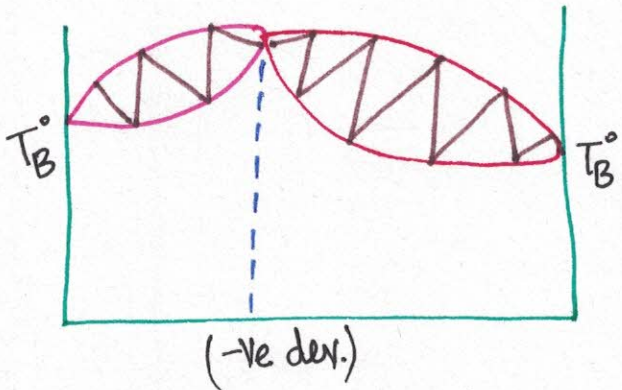
But in case of (+) deviation

Z = BP is same for both and is called azeotropic mixture.



(+) deviation

For (-)ve deviation



In a non ideal soln at a certain compsn, both liquid will have same BP, this compn is called **azeatropic mixture.**

In (+)ve deviation, In this mixture ⟶ Minⁿ BP.
(-)ve deviation, in this mixture ⟶ Maxⁿ BP.

Azeatropic mixture can't be distilled but it is distilled by adding a Particular substance which vapourises the one of the substances.

Q. 100g of sucrose solution in water is cooled to -0.5°C. what of ice would be separated out if the soln starts to freeze at -38°C
$(K_f)_{H_2O} = 1.86\ K\ mol^{-1}\ kg.$

Ans. $0-(.038) = (1.86)(m_1)$

$(molality)_1 = 0.204$

$\therefore [(n)_i]_{solute} = (0.204)(342) = 69.87g$

Therfore, amount of water $= (100-69.87)\,g = 30.13g$

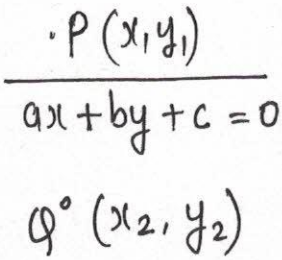
$0-(-0.5) = (1.86)(m_2)$

$(molality)_2 = 0.268$

$\therefore [(n)_F]_{solute} = (0.268)(342) = 91.6\,g$

Therfore, final amount of water $= 8.4\,g$

Amount of ice separated $= (30.13-8.4)\,g$
$= 21.73\,g$



P is above the line;

(b) $(ax_1 + by_1 + c) > 0$

Q is below the line;

(b) $(ax_2 + by_2 + c) > 0$